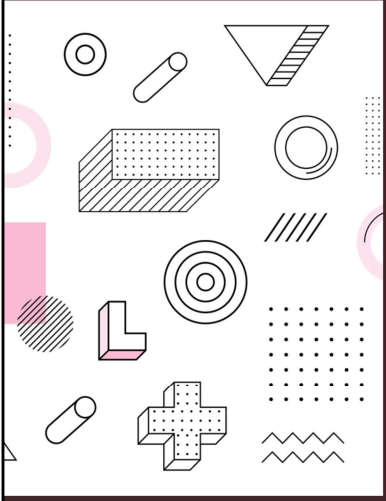


	JOURNAL CLUB #1: REVIEW OF KEY CONCEPTS
	Clipper Young, PharmD, MPH Professor Department of Clinical Sciences & Community Health Fall 2025 - OD3 September 15, 2025 8 - 9:50 am 10:00 - 11:50 am 1:00 - 2:50 pm

person-year

no time

Example/ Case	Person-Time at Risk of Cancer				AIDS Population in Two Cities		
	p	Jan 1, 1990	Jan 1, 2000	Jan 1, 2010	City	A	B
	1	-----X			New Cases	63	22
	2		-----X		Time Period	2005	2005 - 2006
	3	-----D			Population	23,500	8,300
	4	-----L					
	5	-----					
	X = Cancer; D = Died; L = Lost to follow up						
Concept	Incidence Rate (IR)				Cumulative Incidence (CI)		
Formula	IR = (# of new cases of disease) / (person-time at risk in population)				CI = (# of new cases of disease in at-risk population) / (number of people in at-risk population)		
Answer	IR = 2 / 65 PY = 31 / 1,000 PY				A = 63 / 23,500/1 year = 268 / 100,000/1 year B = 22 / 8,300/2 years = 133 / 100,000/1 year		

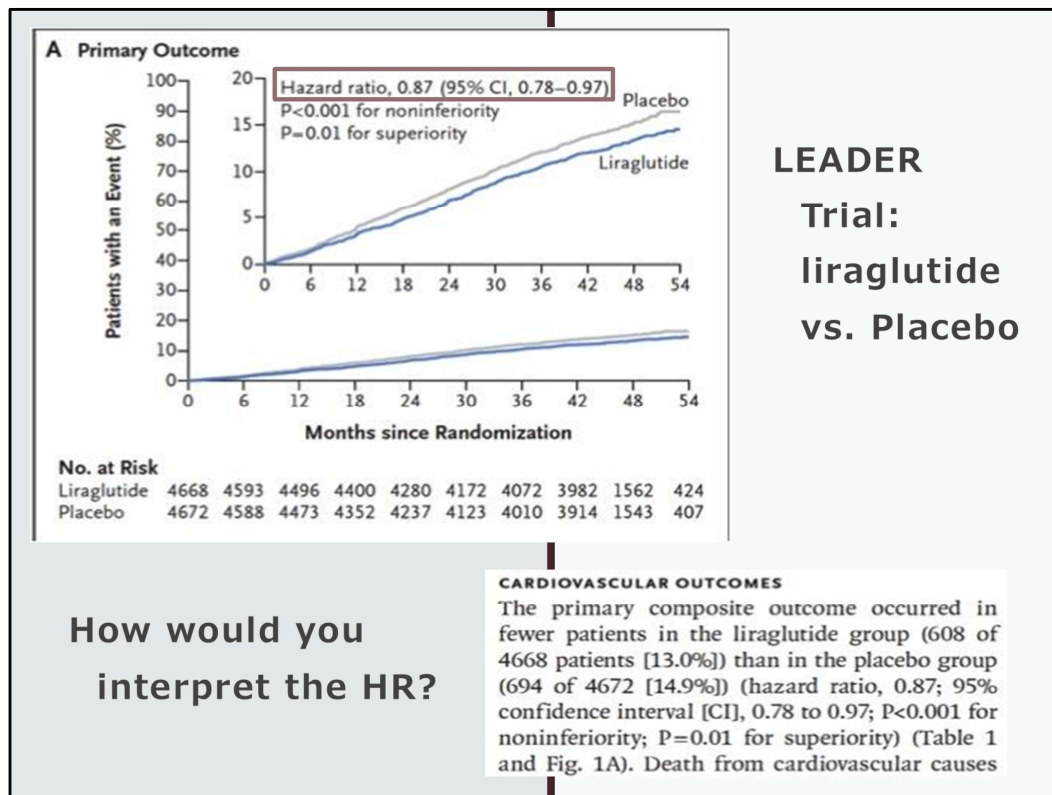
Use one statement to capture the definition of incidence: number of new cases of disease that develop in a population at risk during a specific time frame

people coming in and out (dynamic) - use incidence rate to capture new retention of study

Tends to be used in which study design?	Survival analysis (analysis of death or disease progression; not risk estimation)
How is Hazard Ratio calculated?	$HR = \frac{\text{hazard rate in the treatment group}}{\text{hazard rate in the control group}}$
What is the null value?	1
What is the range?	0 (not likely but possible) to infinity (theoretically)
How would you interpret a Hazard Ratio?	< 1: Defined outcome occurred less frequently in treatment group = 1: No advantage to the treatment (same rate) > 1: Defined outcome occurred more frequently in treatment group

odds ratio, risk ratio/relative risk, hazard ratio (bad outcome, usually safety)

null value: 1 except for rate difference null is 0 because is the difference



reduce CV outcomes

HR = 0.87 reduced likelihood of meeting endpoint

1 - HR = RRR

validity = CI (if the null value is not part of CI = statistical significance)

CV death, non-fatal MI, stroke = composite outcome

study design T2DM risk is higher but doesn't mean that everyone will have a stroke/heart attack - low projected sample size will not have enough power to detect a difference -> why used a composite outcome

Alpha Level

What is it?	Level chosen by investigators as the maximum permissible error in the study
What is the common level chosen in medical research?	0.05 or 5%
How do we apply this level in relation to null hypothesis?	The threshold used for rejecting null hypothesis
Can a smaller alpha level be chosen?	Yes, e.g., 0.01

A smaller alpha level value can be chosen (e.g., 0.01), but this requires more data, more subjects (which means more expense) and/or a larger treatment effect.

[need to justify why changing alpha](#)

p-value and alpha level

- Set an alpha value → perform a statistical test on the data set → generate a p-value (one of the outcomes of a statistical test) → compare p-value with alpha level (two possible outcomes)

Outcome	What Does It Mean?
$p < 0.05$	The null hypothesis is rejected → Null hypothesis is out → There <u>IS</u> a statistically significant difference between the groups
$p \geq 0.05$	The study has failed to reject the null hypothesis → Null hypothesis stands → There is <u>NO</u> statistically significant difference

Example: LEADER Trial

Table 1. Primary and Secondary Outcomes.*

Outcome	Liraglutide (N=4668)	Incidence Rate	Placebo (N=4672)	Incidence Rate	Hazard Ratio (95% CI)	P Value
	<i>no. of patients (%)</i>	<i>no. of events/ 100 patient-yr</i>	<i>no. of patients (%)</i>	<i>no. of events/ 100 patient-yr</i>		
Primary composite outcome†	608 (13.0)	3.4	694 (14.9)	3.9	0.87 (0.78–0.97)	0.01
Expanded composite outcome‡	948 (20.3)	5.3	1062 (22.7)	6.0	0.88 (0.81–0.96)	0.005
Death from any cause	381 (8.2)	2.1	447 (9.6)	2.5	0.85 (0.74–0.97)	0.02
Death from cardiovascular causes	219 (4.7)	1.2	278 (6.0)	1.6	0.78 (0.66–0.93)	0.007
Death from noncardiovascular causes	162 (3.5)	0.9	169 (3.6)	1.0	0.95 (0.77–1.18)	0.66
Myocardial infarction§	292 (6.3)	1.6	339 (7.3)	1.9	0.86 (0.73–1.00)	0.046
Fatal§	17 (0.4)	0.1	28 (0.6)	0.2	0.60 (0.33–1.10)	0.10
Nonfatal	281 (6.0)	1.6	317 (6.8)	1.8	0.88 (0.75–1.03)	0.11
Silent§	62 (1.3)	0.3	76 (1.6)	0.4	0.86 (0.61–1.20)	0.37
Stroke§	173 (3.7)	1.0	199 (4.3)	1.1	0.86 (0.71–1.06)	0.16
Fatal§	16 (0.3)	0.1	25 (0.5)	0.1	0.64 (0.34–1.19)	0.16
Nonfatal	159 (3.4)	0.9	177 (3.8)	1.0	0.89 (0.72–1.11)	0.30
Transient ischemic attack§	48 (1.0)	0.3	60 (1.3)	0.3	0.79 (0.54–1.16)	0.23
Coronary revascularization	405 (8.7)	2.3	441 (9.4)	2.5	0.91 (0.80–1.04)	0.18
Hospitalization for unstable angina pectoris	122 (2.6)	0.7	124 (2.7)	0.7	0.98 (0.76–1.26)	0.87
Hospitalization for heart failure	218 (4.7)	1.2	248 (5.3)	1.4	0.87 (0.73–1.05)	0.14
Microvascular event	355 (7.6)	2.0	416 (8.9)	2.3	0.84 (0.73–0.97)	0.02
Retinopathy	106 (2.3)	0.6	92 (2.0)	0.5	1.15 (0.87–1.52)	0.33
Nephropathy	268 (5.7)	1.5	337 (7.2)	1.9	0.78 (0.67–0.92)	0.003

* Hazard ratios and P values were estimated with the use of a Cox proportional-hazards model with treatment as a covariate.

† The primary composite outcome in the time-to-event analysis consisted of the first occurrence of death from cardiovascular causes (181 patients in the liraglutide group vs. 227 in the placebo group), nonfatal (including silent) myocardial infarction (275 vs. 304), or nonfatal stroke (152 vs. 163). The P value is for superiority.

‡ The expanded composite outcome included death from cardiovascular causes, nonfatal myocardial infarction, nonfatal stroke, coronary revascularization, or hospitalization for unstable angina pectoris or heart failure.

§ This analysis was not prespecified.

Marso SP, et al. (2016) Liraglutide and Cardiovascular Outcome in Type 2 Diabetes. N ENGL J MED 375; 4.

P - T2DM

I - liraglutide

C - comparison, placebo

O - primary composite outcome (CV event, non fatal MI, s

Type of q - prevention, therapeutic

Type of study - randomized controlled trial

measure of disease freq: HR

Different Levels of p-values

p-value	Meaning
≥ 0.05	Not statistically significant
< 0.05	> 95% probability that results did not happen by chance; at most 4.999...% chance that results happened at random (researchers willing to take that risk)
< 0.01	> 99% probability that results did not happen by chance; at most 0.999...% chance that results happened at random
< 0.001	> 99.9% probability that results did not happen by chance; at most 0.099...% chance that results happened at random

Confidence Interval

What is it?	Provides the same significance information as p-value plus precision of results
What is the relationship between CI and alpha level?	$CI = 1 - \alpha$
What does a narrow CI imply?	Narrow CI = high precision Wide CI = low/poor precision

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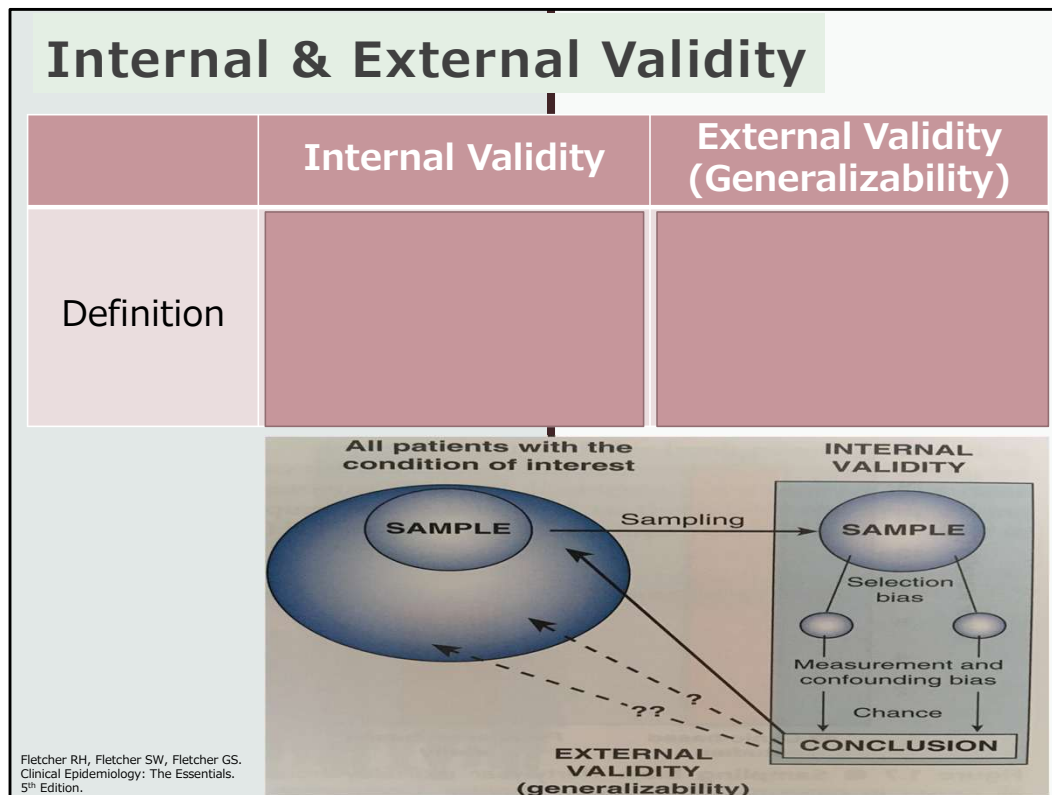
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how would you translate the primary composite outcome in table 1?



Internal validity:

The study was done correctly and its results reflect “truth” within constraints of chance

in order to answer if there is internal validity?

1. observation
2. is the question testable
3. controlling biases within model
4. data interpretation - need power analysis bc will determine sample size to see if a difference was a
5. did the conclusion answer the question?

right study design and science

Study Type	Key Features/Definition	Advantages	Disadvantages
Descriptive Studies			
Case Report			
Case Series			
Cross-Sectional			
Ecological (Correlational)			
Analytic Studies			
Experimental Studies			
Clinical Trials			
Observational Studies			
Cohort Studies			
Prospective Cohort Studies			
Retrospective Cohort Studies			
Case-Control			

Clinical Trials

Clinical Trial – Definition/ Key Features

- Research protocol, not the subject, determines **exposure/treatment** status
 - the outcome has **not** occurred at the time of classification/randomization
- Subjects are followed forward in calendar time to determine **outcome** status

no intervention in observational
RCT - assigning intervention

CLINICAL TRIAL = EXPERIMENT

Advantages/Strengths

1. Gold standard (options to **minimize** bias = Randomization, Controlled (Comparison group), also Blinding (Single or Double or Triple))
2. Can study multiple exposures
3. Calculate incidence rates
4. Better data quality
5. Can collect information on potential confounders

Disadvantages/Limitations

1. Need equipoise
2. Ethical considerations
3. Can only study relatively common diseases/outcomes
4. Requires large sample
5. Longer study duration
6. Risk of loss to follow-up
7. Expensive
8. External generalizability may be limited

2 active ingredients in a study - head to head study
equipoise = no knowledge if the intervention will produce help c

Screening test terms:

sensitivity
specificity
PPV
NPV

sensitivity and specificity - independent of prevalence
PPV and NPV as prevalence increases PPV increases
NPV will decrease slightly (can assume the same)